Grade 7 Accelerated
Unit 1
Lesson 6 Practice

Name_Answer Key_________
Date_____________________
Period___________________

A realtor earns a base salary of $\$30{,}000$ a year plus $3\%$ of her yearly sales. If Victoria wants to make a $\$60{,}000$ annual salary this year, how much will she need to sell?

1. What are the known quantities?

   - $\$30{,}000$ base salary
   - additional earnings of $3\%$ of sales
   - $\$60{,}000$ goal salary

2. What is the unknown quantity?

   how much the realtor sells in a year

3. Write an equation that could be used to find the unknown quantity using the variable you defined.

   $0.03x + 30{,}000 = 60{,}000$

4. Solve the equation.

   $0.03x + 30{,}000 = 60{,}000$

   $\frac{0.03x}{0.03} = \frac{30{,}000}{0.03}$

   $x = 1{,}000{,}000$

5. Explain what the solution means in this context.

   The realtor would have to sell $\$1{,}000{,}000$ in property to make $\$60{,}000$ for the year.

Fiona has a coupon for $\$3$ off the price of her meal at a local fast food establishment. Her favorite meal is on sale at a special rate of $10\%$ off the combo price. If Fiona paid $\$8.25$ for her meal, before taxes, what was the original price of the meal?

6. What are the known quantities?

    - $\$3$ off coupon
    - $10\%$ off combo price
    - $\$8.25$ for the meal

7. What is the unknown quantity?

    original price of the combo meal

8. Write an equation that could be used to find the unknown quantity using the variable you defined.

    $0.9x - 3 = 8.25$

9. Solve the equation.

    $0.9x - 3 = 8.25$

    $\frac{0.9x}{0.9} = \frac{11.25}{0.9}$

    $x = 12.5$

10. Explain what the solution means in this context.

    The original price of Fiona's favorite meal is $\$12.50$.